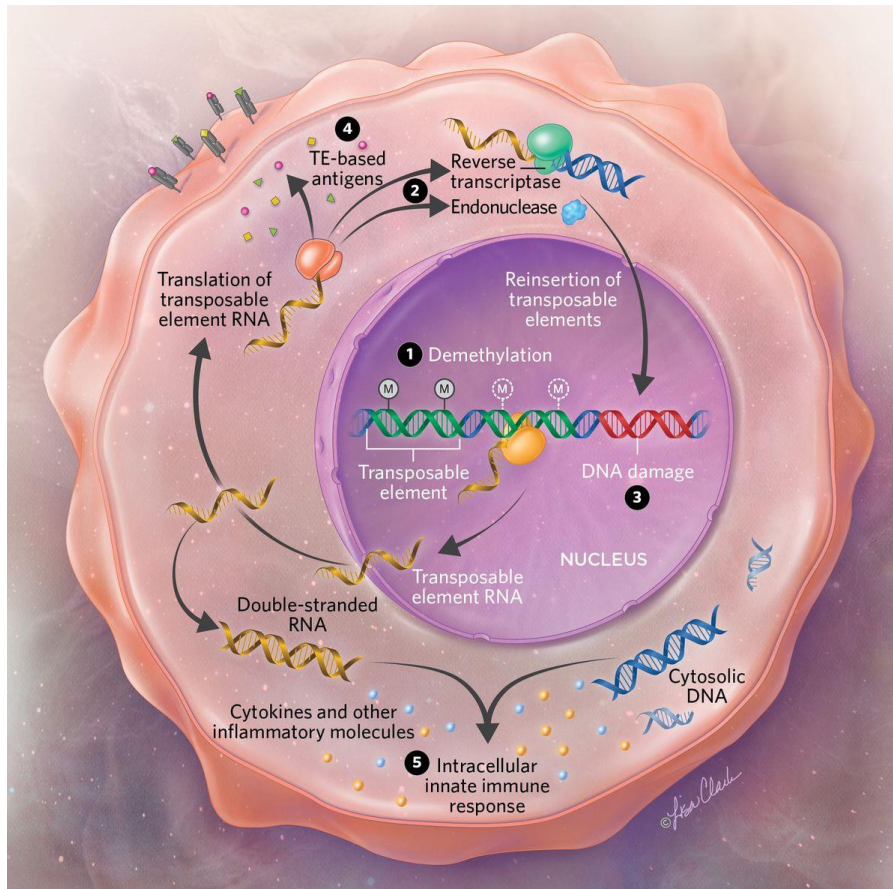


Transposable elements in human genetic disease

Transposable element insertions have been known to cause human genetic disease since the 1980s.

The scope of their contributions to heritable phenotypes is now starting to be uncovered.

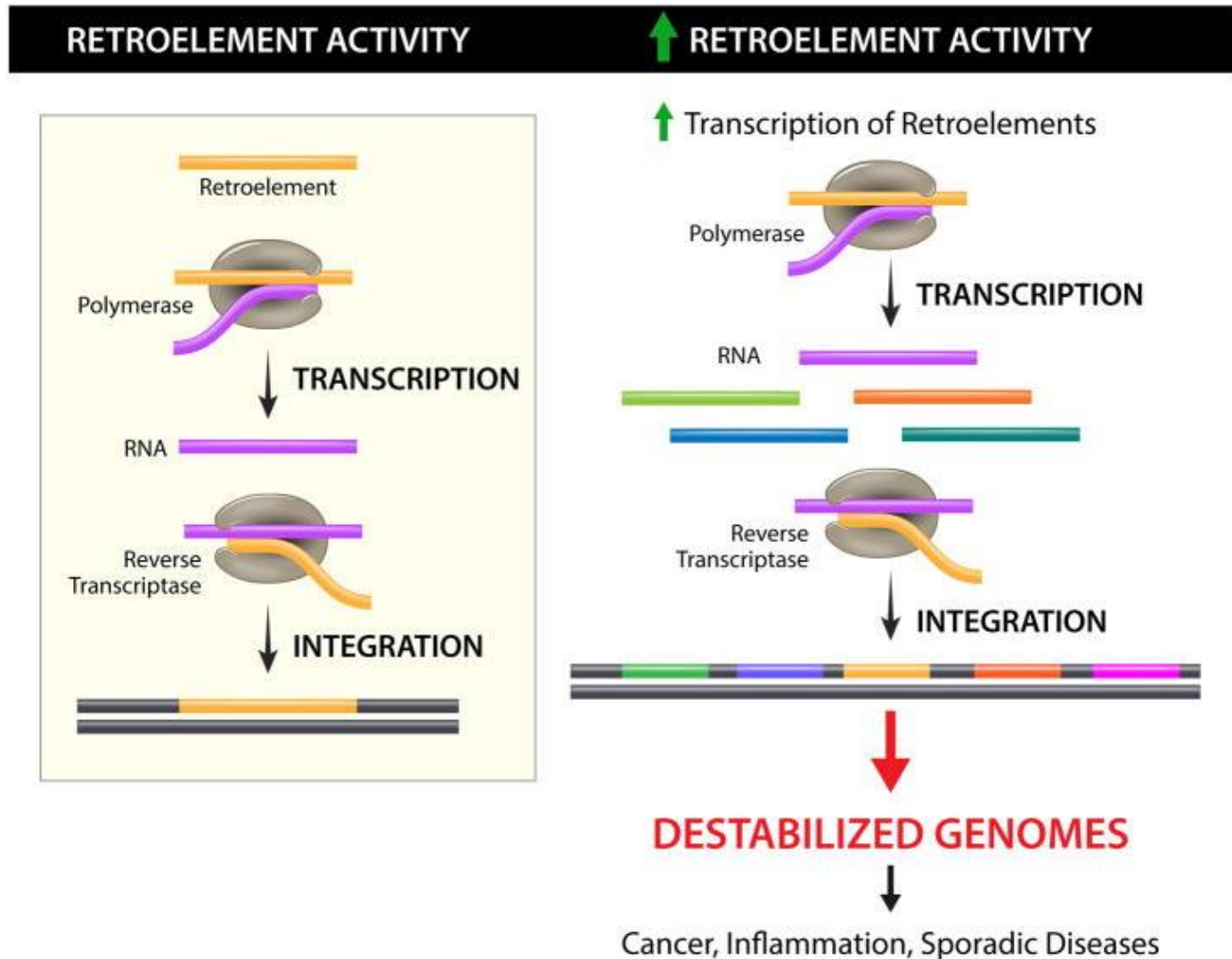


Transposable elements are abundant in the human genome, and great strides have been made in pinpointing variations in these repetitive sequences using whole-genome sequencing. **Now, the focus is shifting to understanding their expression and regulation**, and the functional consequences of their insertion and retention in the genome over time. Whereas transposable element insertions have been known to cause human genetic disease since the 1980s, the scope of their contributions to heritable phenotypes is now starting to be uncovered. Here, we review the many ways human retrotransposons contribute to genome function, **their dysregulation in diseases including cancer** and how they affect genetic disease.

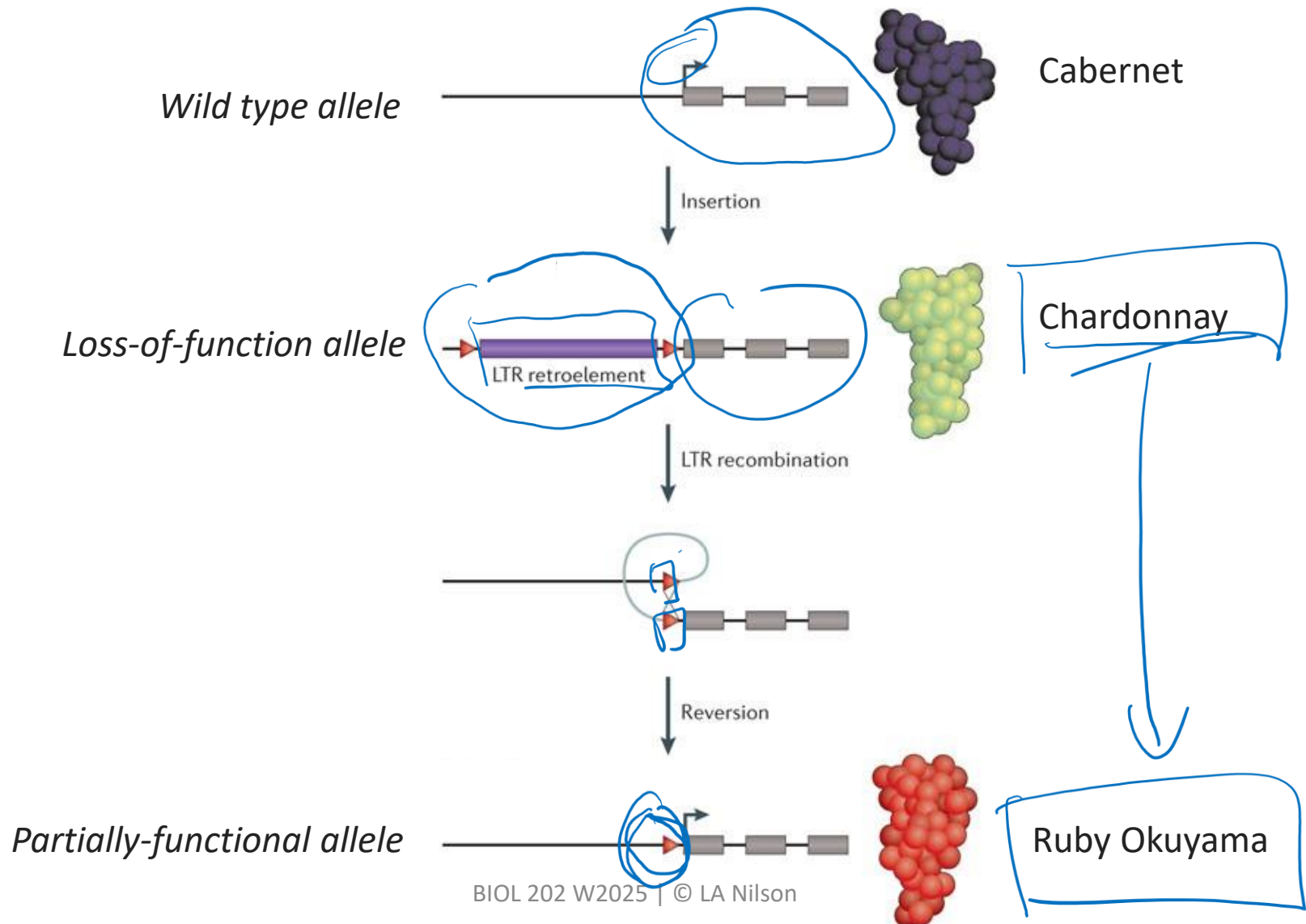
The Dynamic Genome

- Discovery of transposable elements through study of “unstable” alleles ✓
- Impact of transposable elements on the genome ✓
- Genomic surveillance mechanisms that limit transposon mobility]

Transposable element mobility can destabilize genomes



Transposable elements as a potential a source of major phenotypic novelty and adaptive change



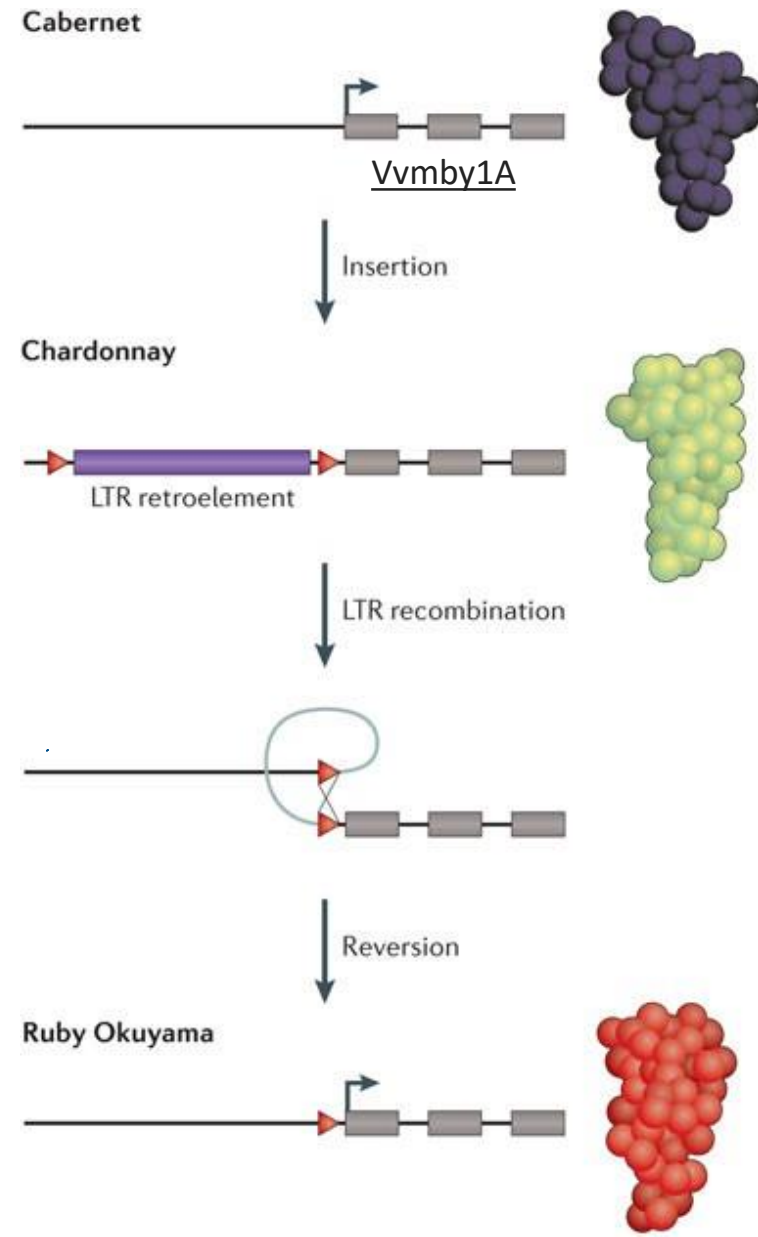
Transposable elements as a source of phenotypic novelty

The *Vvmyb1A* gene is required for production of purple pigment in Cabernet grapes.

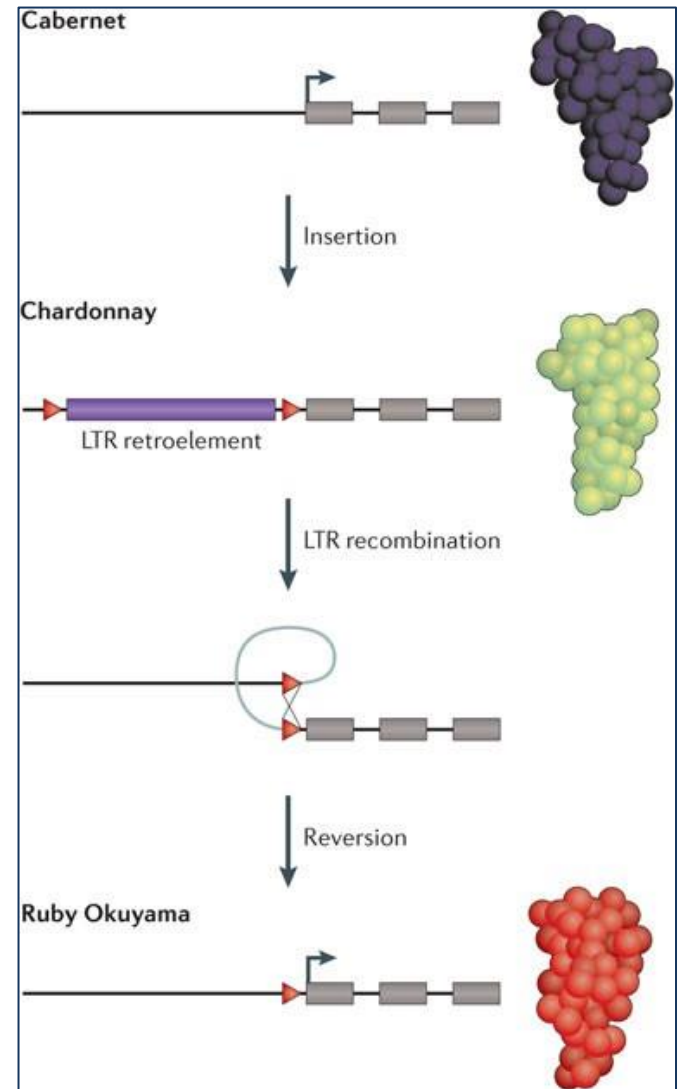
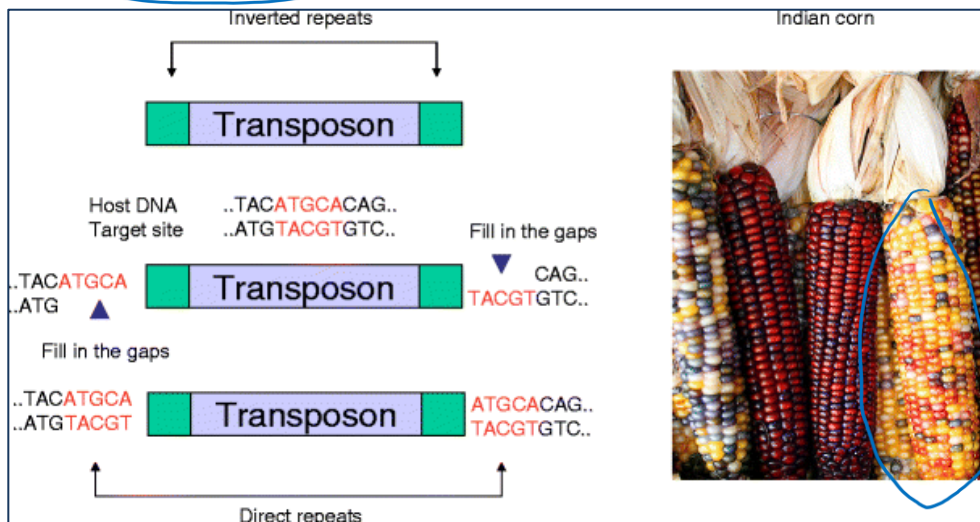
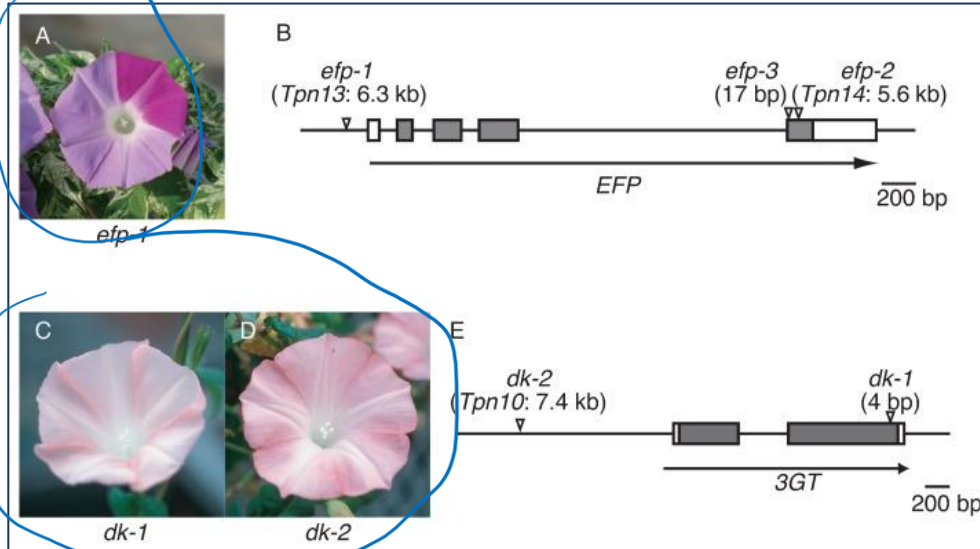
An initial insertion of a *Gret1* long terminal repeat (LTR) retrotransposon resulted in a loss-of-function allele of the *Vvmyb1A* gene, leading to a loss of colour in the fruit of the Chardonnay variety.

A subsequent rearrangement in *Gret1* results in revertant, coloured grapes in varieties such as Ruby Okuyama.

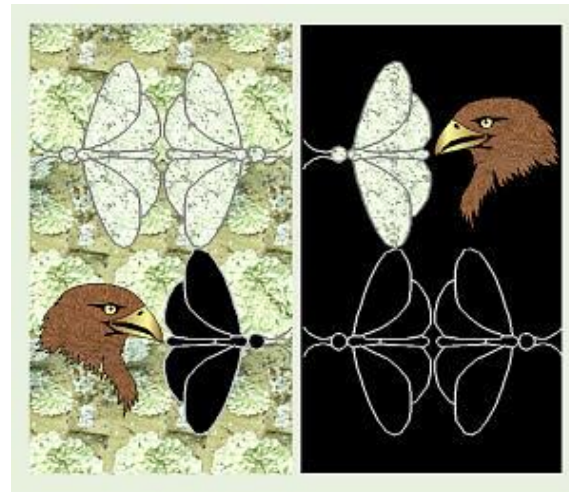
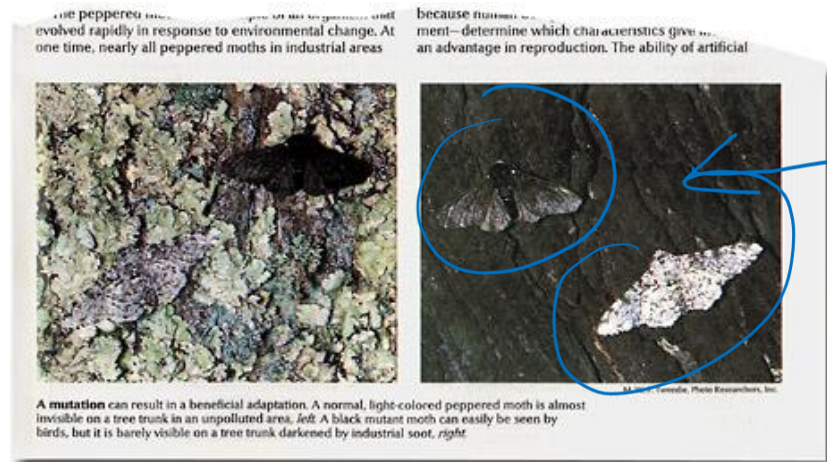
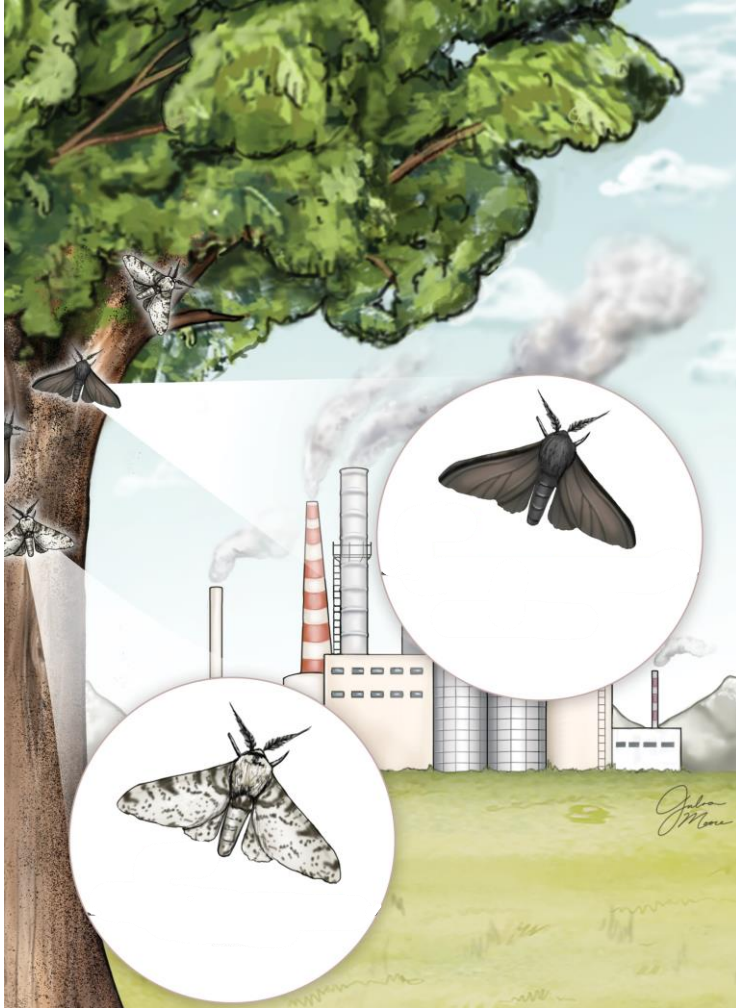
Exons are depicted as grey boxes. The LTRs flanking Gret1 just upstream of the Vvmyb1A gene are depicted as red triangles.



Transposable elements as a potential source of major phenotypic novelty

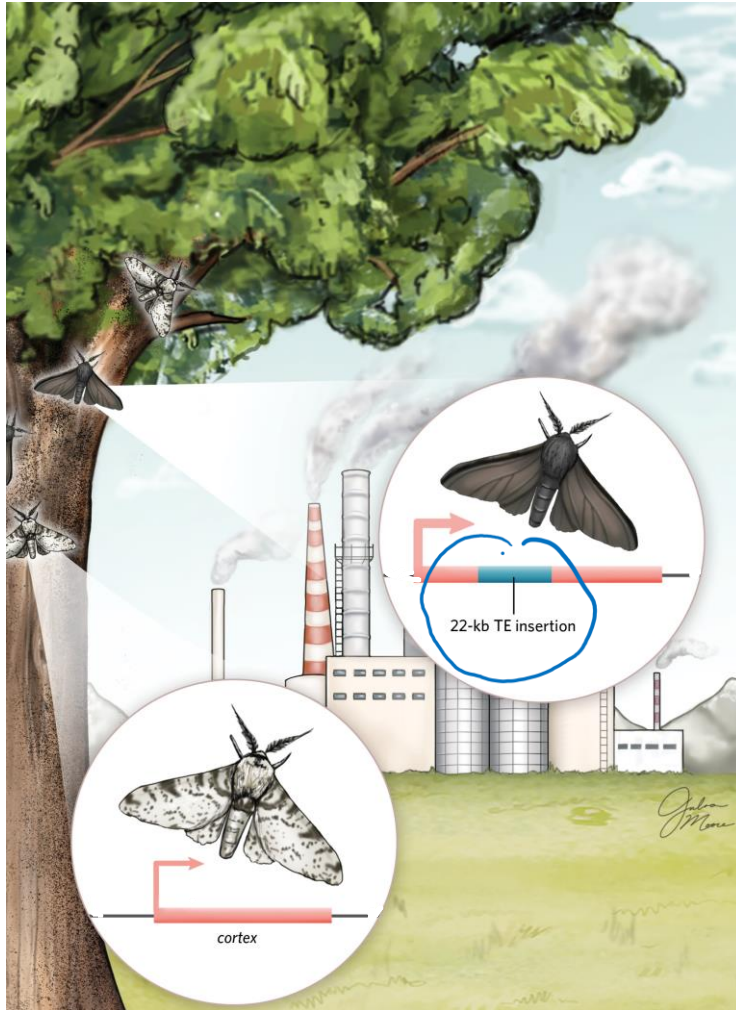


Transposable elements as a potential source of major phenotypic novelty and adaptive change



[Peppered Moths | Natural Selection Game \(asu.edu\)](http://asu.edu)

Transposable elements as a potential a source of major phenotypic novelty and adaptive change



(excerpt from scientific article in the journal Nature, in 2016 – see link below)

“Here we show that the mutation event giving rise to industrial melanism in Britain was the insertion of a large, tandemly repeated, transposable element into the first intron of the gene cortex. Statistical inference based on the distribution of recombined carbonaria haplotypes indicates that this transposition event occurred around 1819, consistent with the historical record.”

“The discovery that the mutation itself is a transposable element will stimulate further debate about the importance of 'jumping genes' as a source of major phenotypic novelty.”

Genome Surveillance

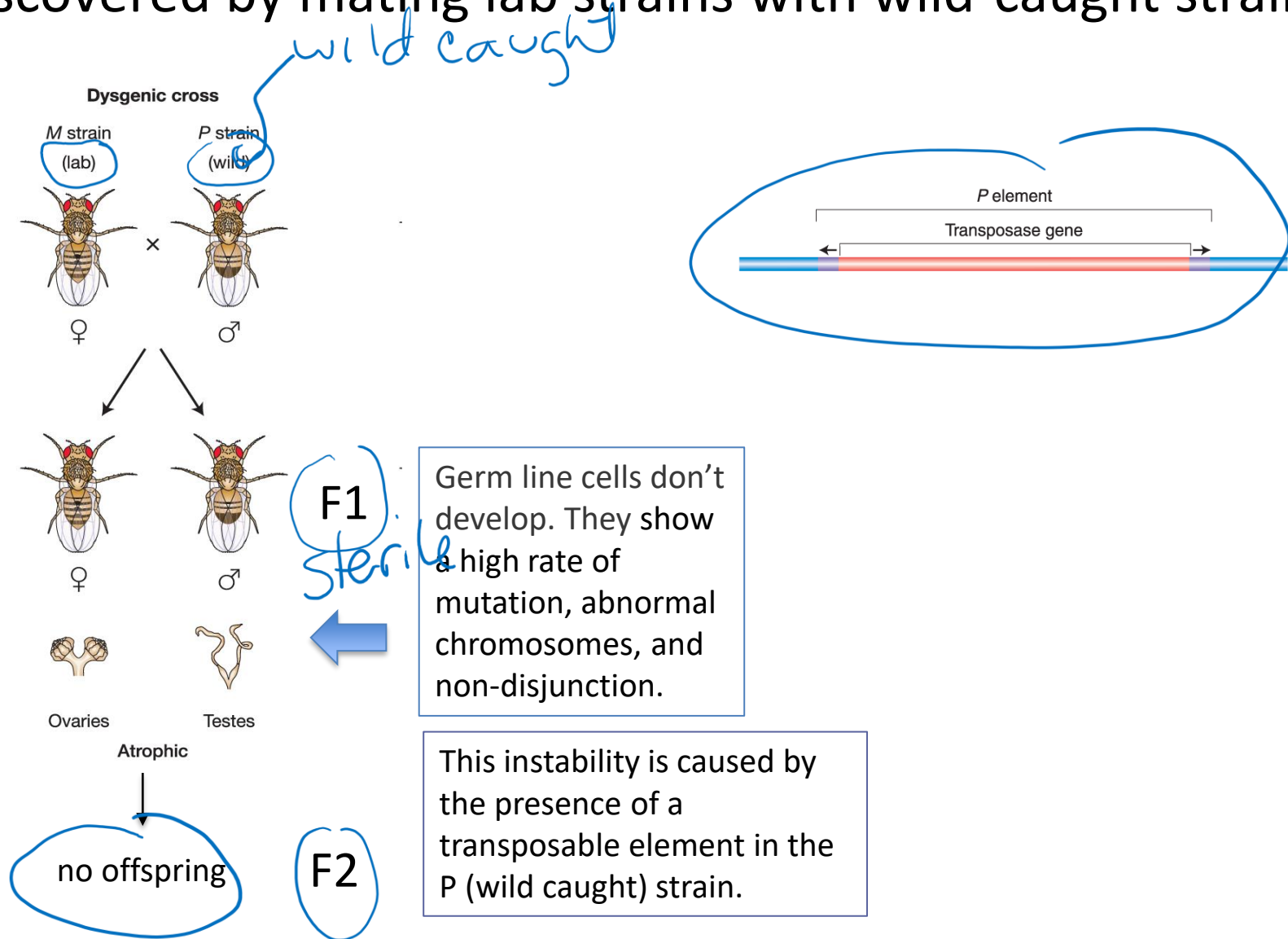
Limiting the impact of transposable element mobility

Two general points to keep in mind:

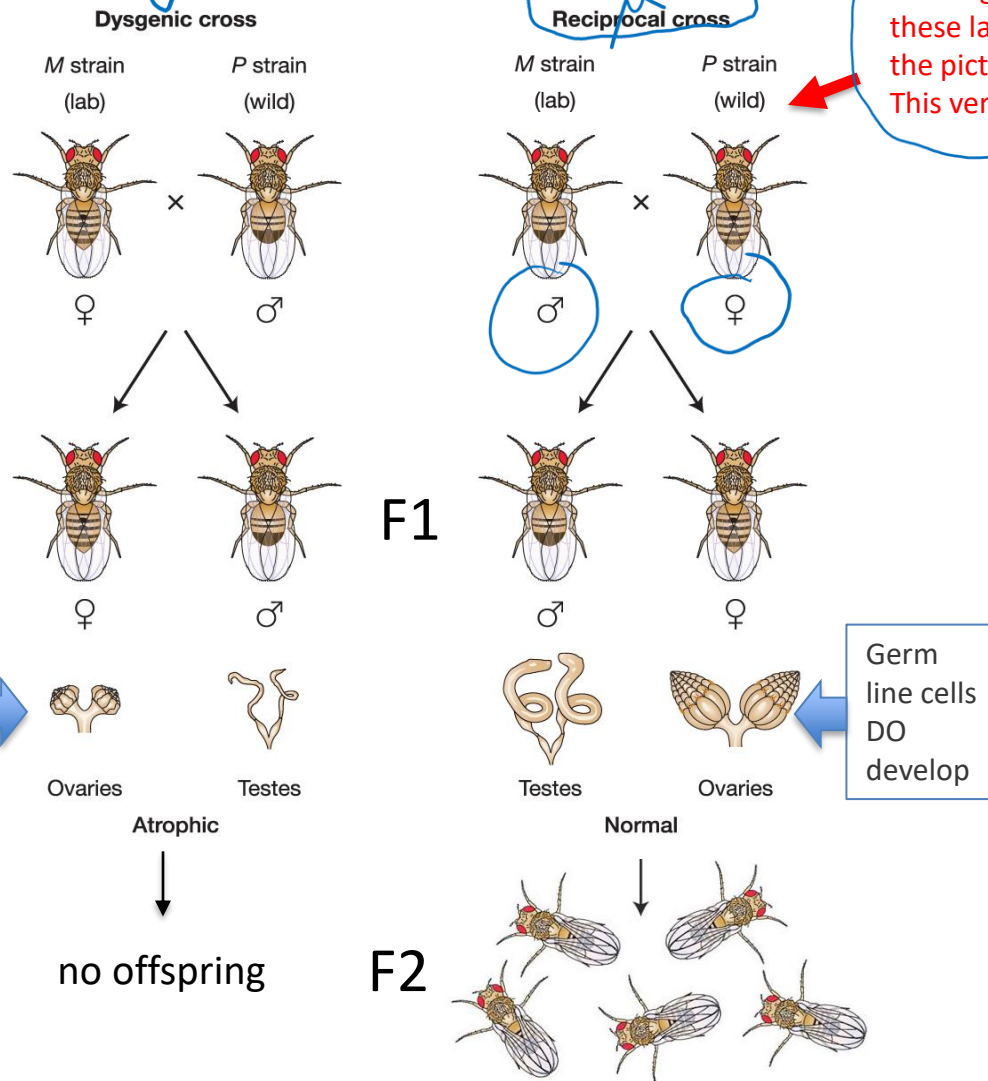
- Transposable element mobility leads to mutations and genome instability. ✓
- Cells have mechanisms to repress transposable element mobility and protect genome integrity. ✓

Also: understand the experiments, results, and interpretations ✓

The Drosophila P-element (a type 1 transposon) was discovered by mating lab strains with wild-caught strains



Drosophila P-elements were discovered by mating lab strains with wild-caught strains



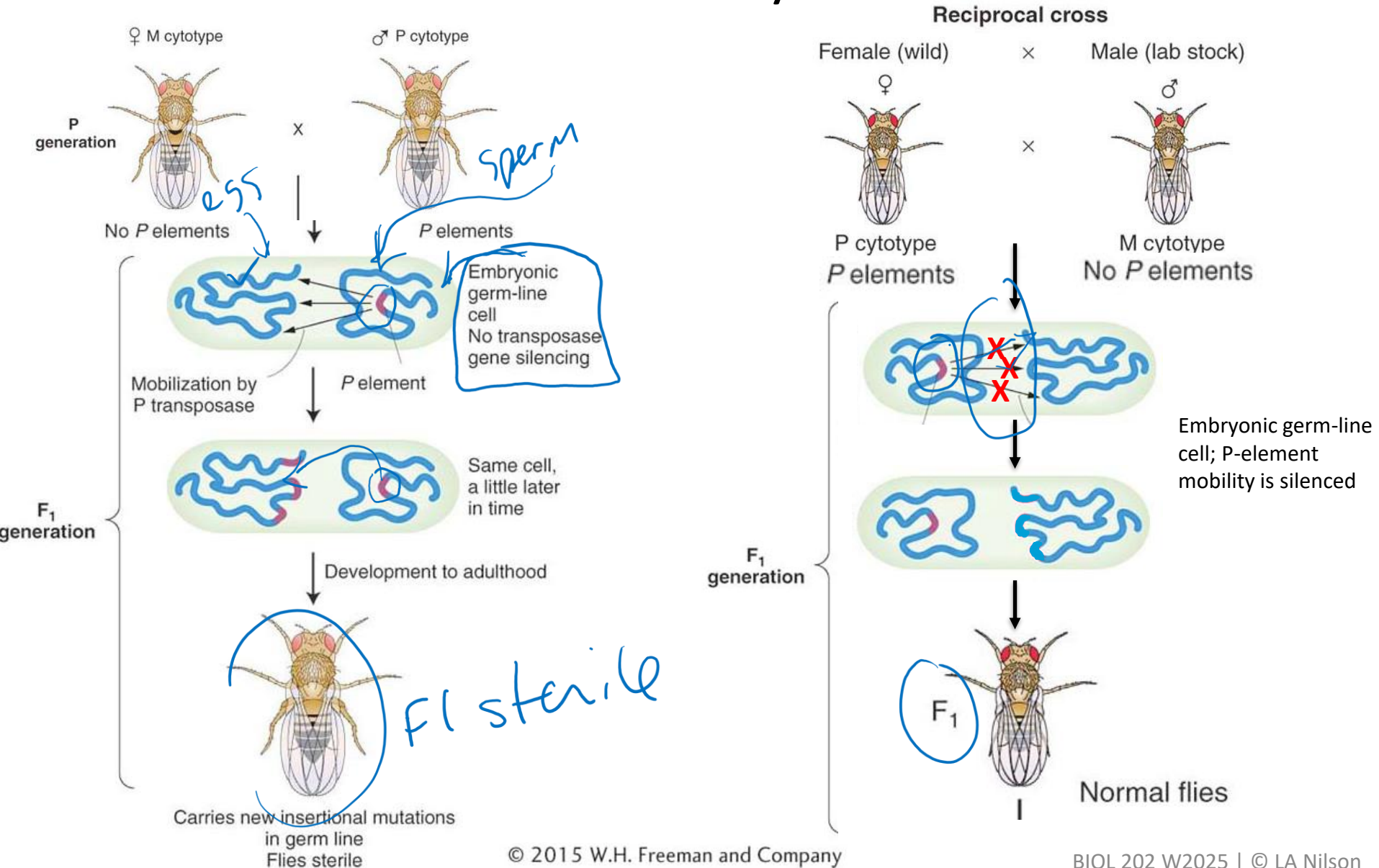
warning:: there is a mistake in these labels in the version of the picture in the textbook. This version here is correct.

If the P strain contains transposable elements, why does this cross yield normal F2 progeny?

Observation: In this cross, the genome of the germ cells remains intact despite the presence of P elements.

Interpretation: A mechanism must exist that preserves genome integrity, perhaps by blocking P-element transposon mobility

Mechanisms exist that repress transposable element mobility



Cells have mechanisms that repress transposable element mobility.

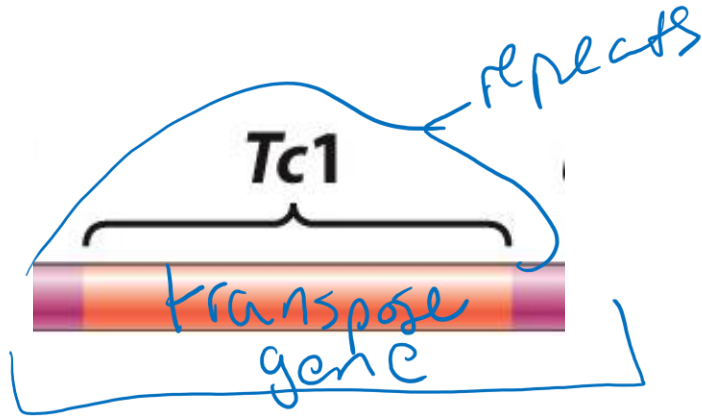
What are they and how do they work?

How does a cell silence its transposons?

What is the molecular mechanism of transposon silencing?

How do we figure this out?

C. elegans also have transposable elements, plus mechanisms for repressing transposable element mobility



C. elegans Tc1 element

an autonomous DNA transposon, analogous to the maize Ac element, and the *Drosophila* P element.

32 Tc1 element copies in the *C. elegans* genome



Observation: Tc1 elements transpose in somatic cells but not germline cells.

Interpretation: The transposase gene in each Tc1 copy is silenced in germline cells.

Hypothesis: The germline cells contain some kind of "silencing machinery" that represses Tc1 mobility. **The components of this "machinery" are encoded by genes.**

How do we identify the genes that cells use to repress transposable element mobility?

Rationale:

- If we knock out a gene involved in repressing transposable element mobility*, then transposable elements will be able to move.
- Conversely, if we find evidence that a mutant strain shows evidence of transposable element mobility, then we can infer that the mutated gene is required to repress transposable element mobility.
- Figuring out which gene is mutated in such a strain will identify a part of the cellular machinery that represses transposable element mobility.

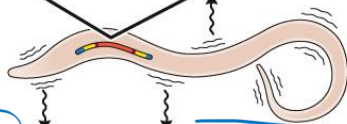
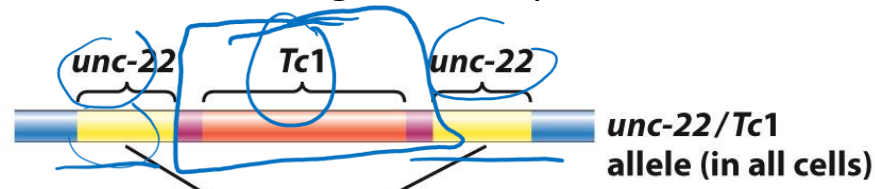
*Q: If we don't know how the process works, how do we know which genes to knock out?

A: We don't. We knock them out at random, and see which ones allow transposable elements to move. We then infer that the knocked-out genes are involved in silencing of Tc1 mobility. This is called a "genetic screen."

If transposon mobility is no longer repressed, Tc1 can jump out of the *unc-22* gene, restoring *unc-22* function

Tc1 insertion in the *unc-22* gene disrupts *unc-22* function

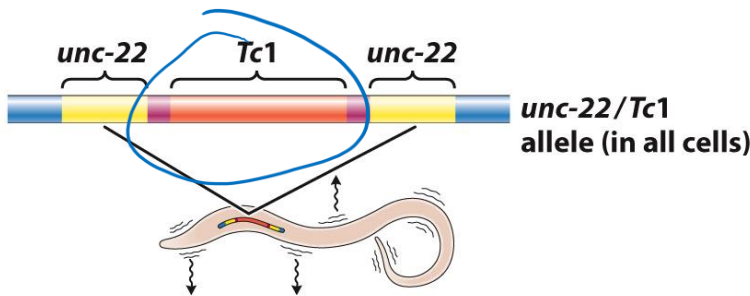
Approach: Do a "Genetic Screen" to identify mutations that allow Tc1 to mobilize. We then infer that the knocked-out genes are involved in **silencing** of Tc1 mobility.



Twitching/uncoordinated worm

Derivative mutant strains

If Tc1 does not mobilize ("jump")



Twitching/uncoordinated worm

If Tc1 mobilizes ("jumps")



Smooth-gliding worm

germline
★
restored!

F1

F1

Genetic screen: mutate genes at random, then select those in which *unc-22* function is restored.

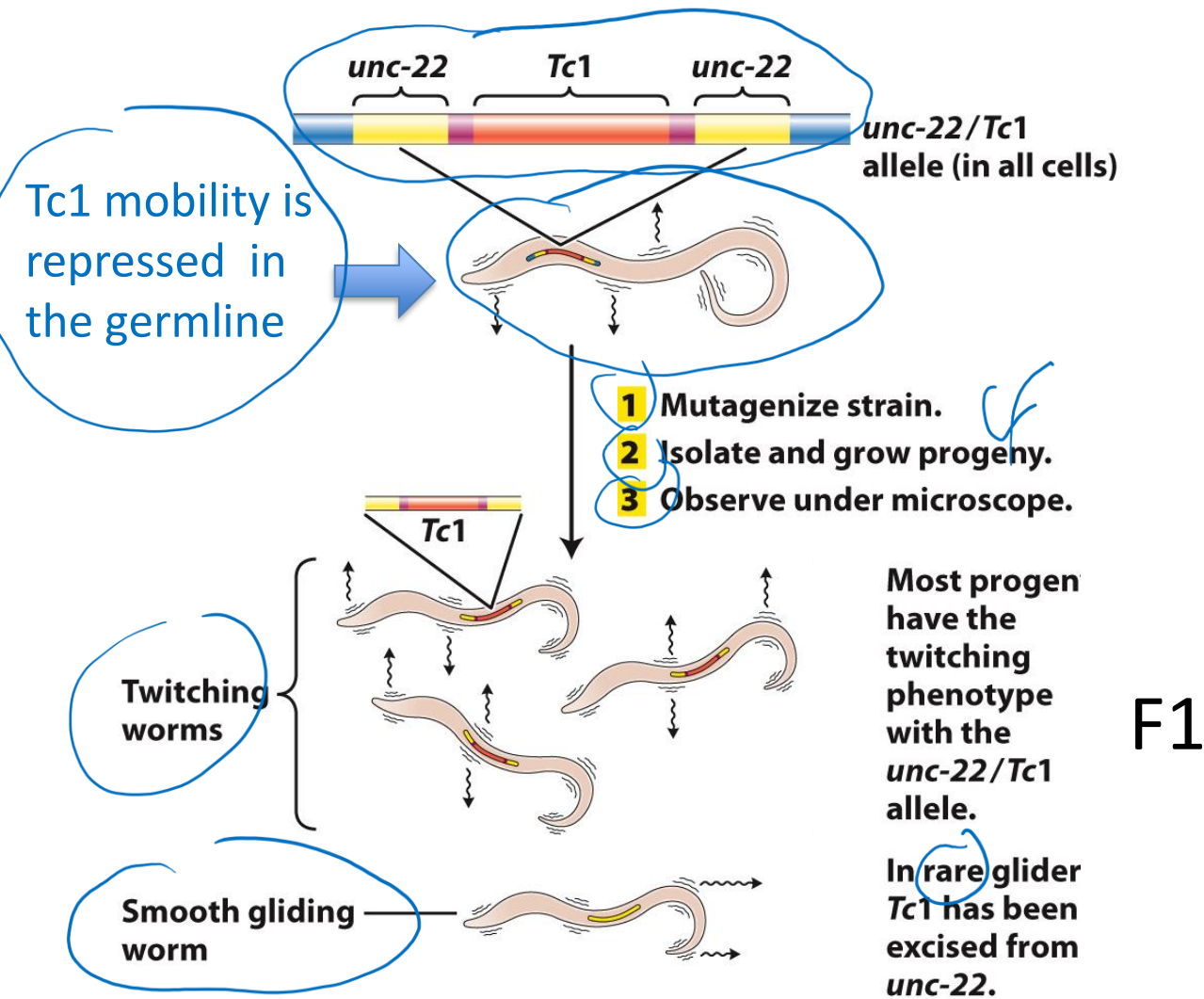


Figure 15-25
Introduction to Genetic Analysis, Eleventh Edition
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How do we identify the genes that cells use to repress transposable element mobility?

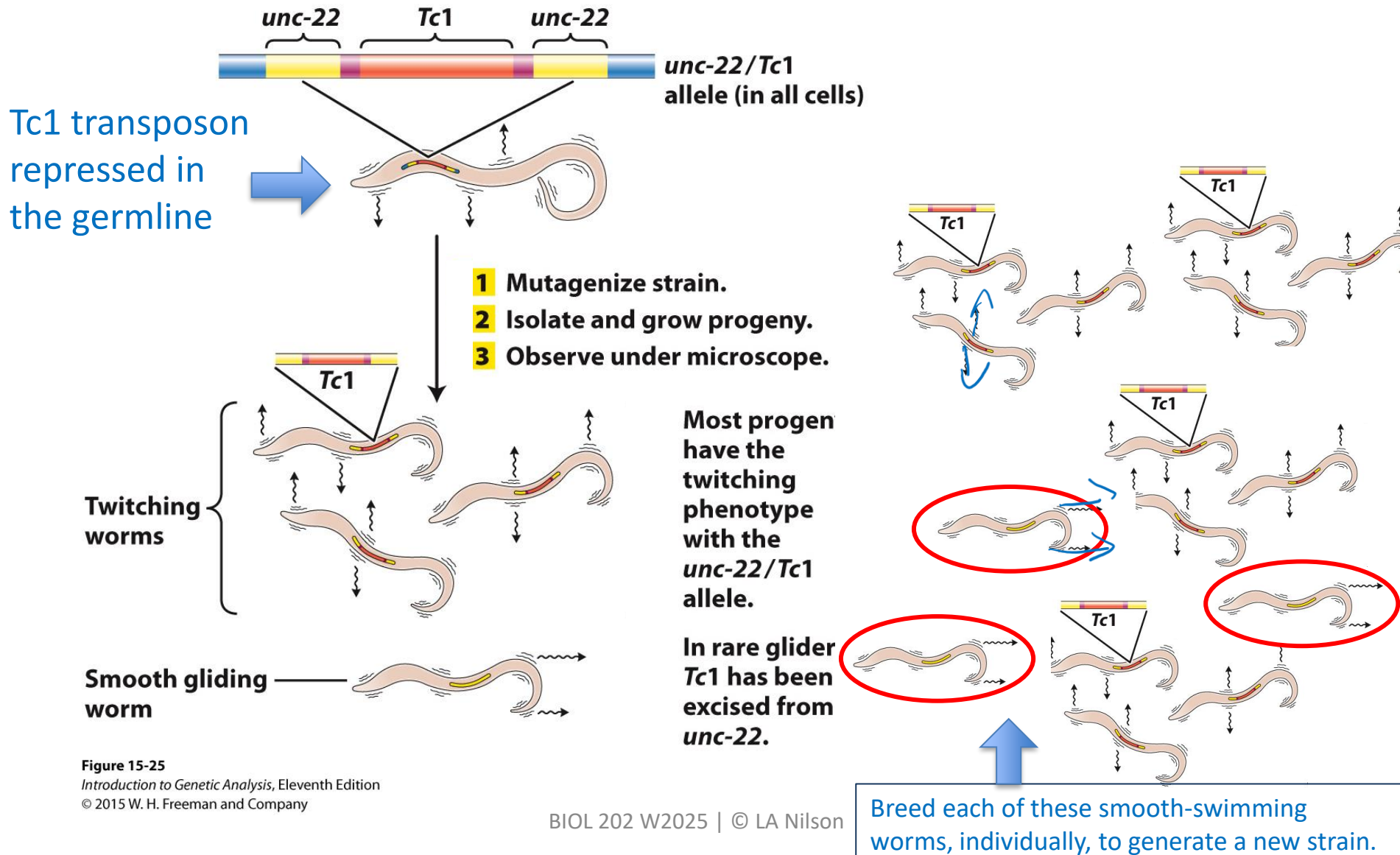


Figure 15-25
Introduction to Genetic Analysis, Eleventh Edition
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Think Break

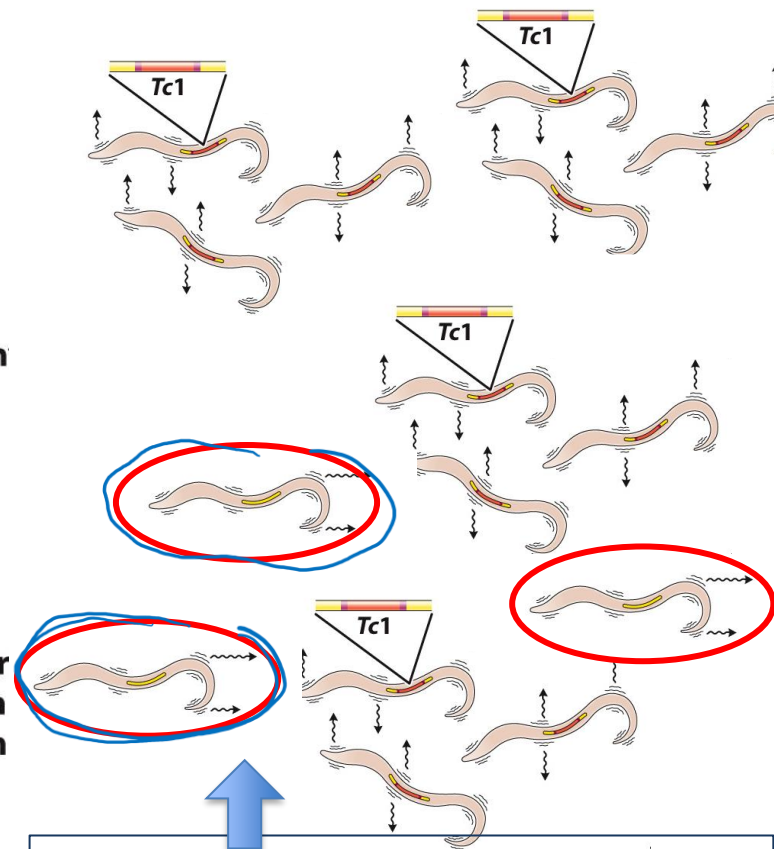
What happened to these worms?

- A. Their *unc-22* gene was knocked out. F
- B. Their *unc-22* gene reverted to wild type. T
- C. The Tc1 element was silenced. F
- D. The Tc1 element was mobilized. T
- E. The mutagen knocked out the Tc1 element. T/F
- F. The mutagen knocked out another gene. T

Answers: B and D and F;

Most progen have the twitching phenotype with the *unc-22/Tc1* allele.

In rare glider Tc1 has been excised from *unc-22*.



Breed each of these smooth-swimming worms, individually, to generate a new strain.

What happened to these worms?

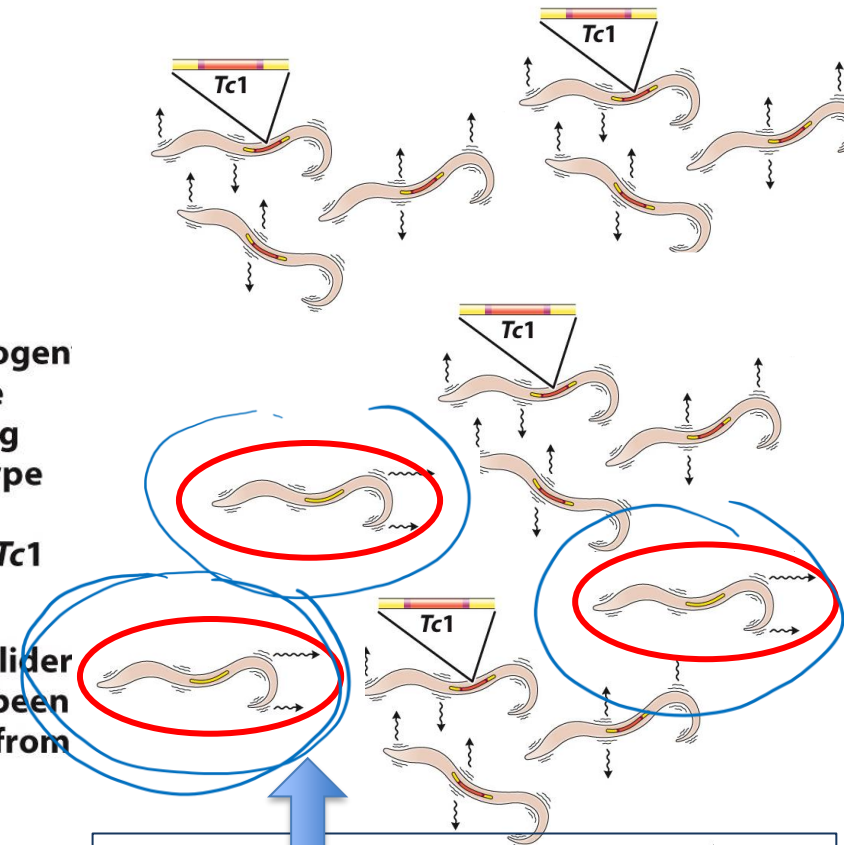
Is the same gene knocked out
each worm?

How would you tell?

complementation
test

Most progen
have the
twitching
phenotype
with the
unc-22/Tc1
allele.

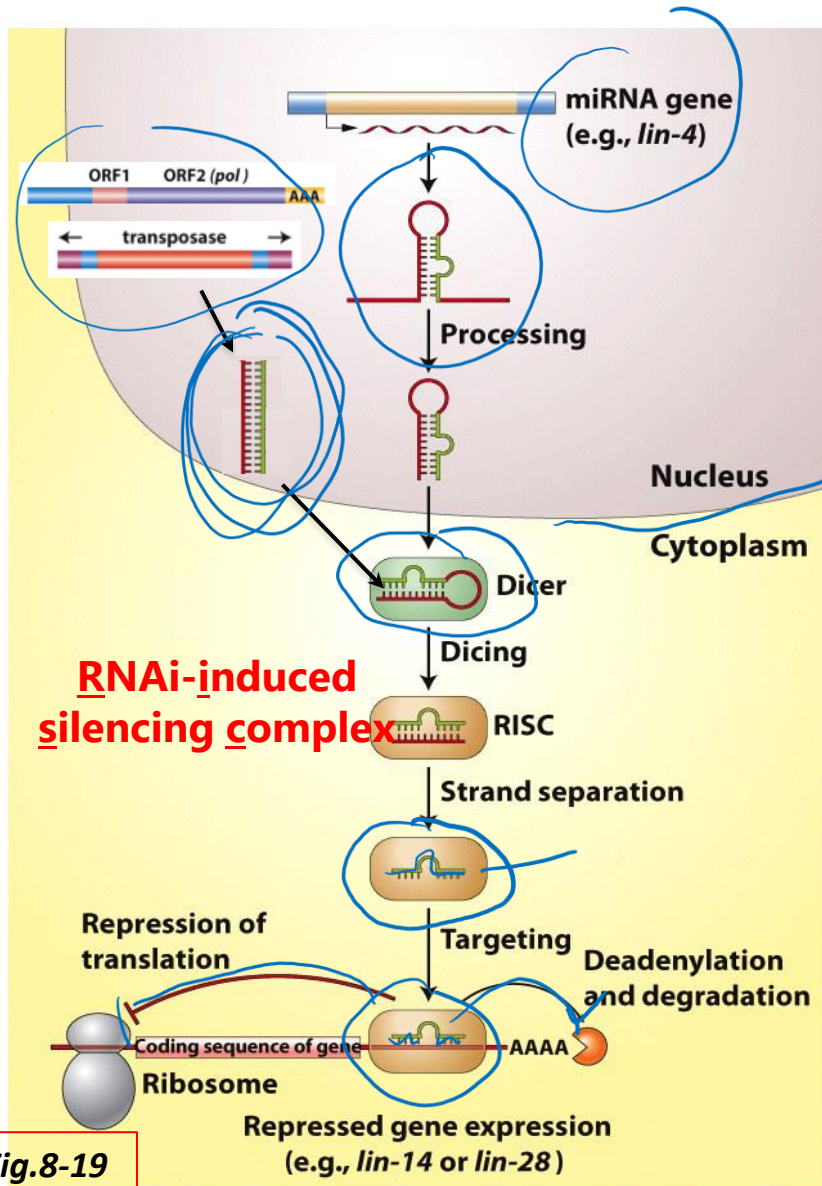
In rare glider
Tc1 has been
excised from
unc-22.



Breed each of these smooth-swimming
worms, individually, to generate a new strain.

Screens identify 25 genes that block Tc1 mobility.

Many of these genes are involved in the RNAi silencing pathway.



Based on short (~21-25 nt) RNAs derived from double-stranded RNA precursors

Processing involves many steps, including cleavage by Dicer

Once fully processed, miRNAs bind to RISC complex

miRNA/RISC complex recognizes mRNA targets through sequence complementarity, and either promotes their degradation or represses their translation

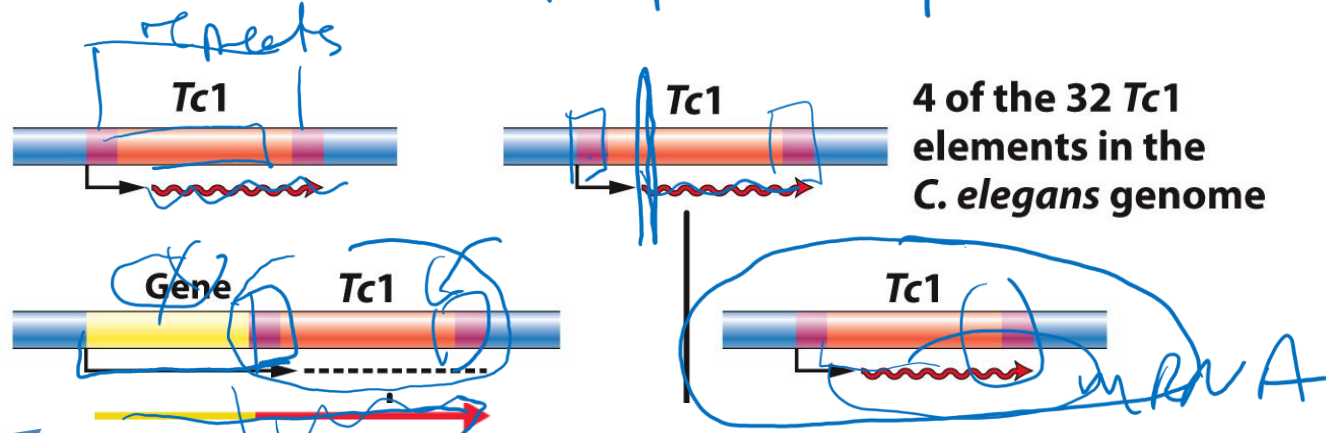
microRNA (**miRNA**): endogenous gene

siRNA: foreign gene (transgene, virus, TE)

Fig.8-19

A single Tc1 can repress all Tc1 mobility in a cell

led to repression of

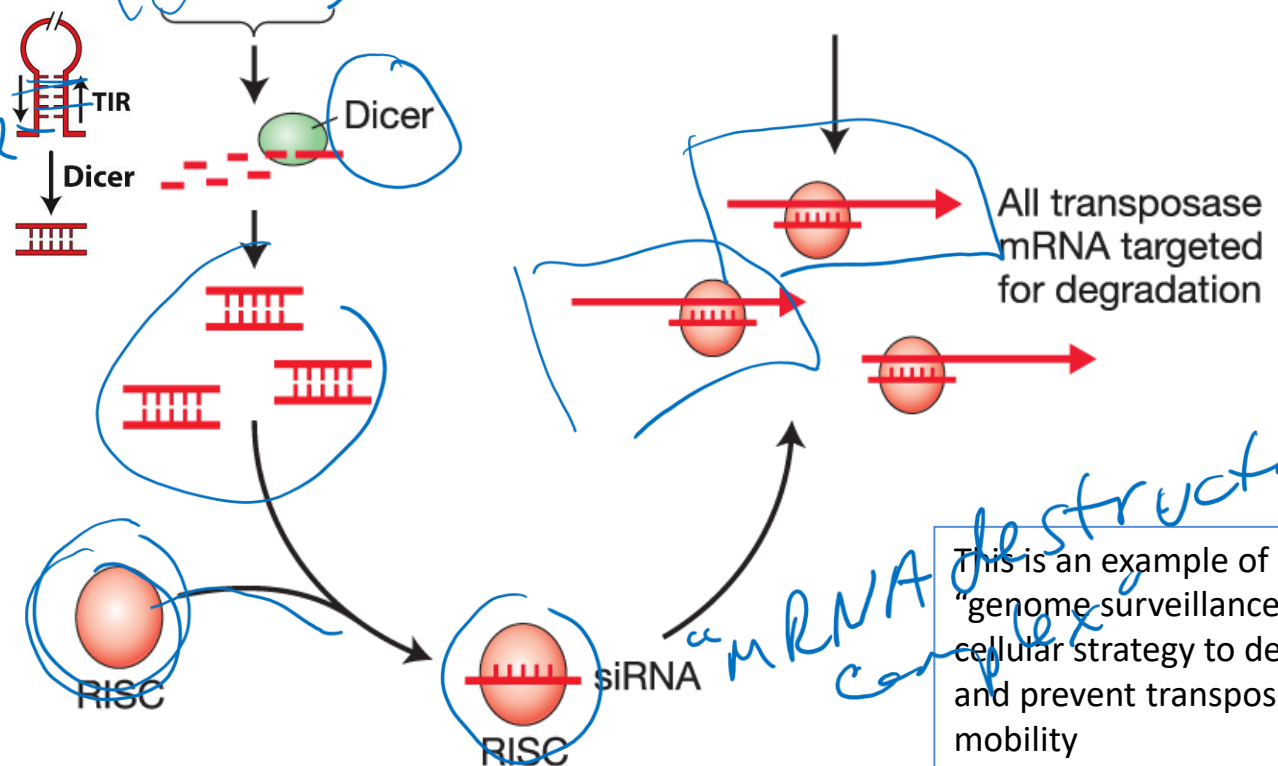


“read-through transcription from a Tc1 transposon that happened to insert next to a cellular gene.”

transcribed into Tc1 RNA

Both ends of Tc1 are transcribed. These contain 54-bp terminal inverted repeats (TIRS)...

... so the Tc1 RNA spontaneously forms dsRNA



This is an example of “genome surveillance” – a cellular strategy to detect and prevent transposon mobility

mRNA destruction complex